JUBIS MARKETING ERP — TECHNICAL & FUNCTIONAL INVENTORY

Document: Full System Specification & QA Blueprint **Generated:** October 2026 **Status:** Production Verified

1. System Architecture & Tech Stack

1.1 Frontend & User Interface

- **Core Framework:** Vue 3 (v3.4.0) with Composition API (`<script setup>`) and Single-File Components.
- **Client-Server Bridge:** Inertia.js Vue 3 Adapter (v2.0.0) with Ziggy (v2.0) for Laravel named route mapping.
- **Styling & Design System:** Tailwind CSS (v3.2.1), @tailwindcss/forms (v0.5.3), PostCSS (v8.4.31), Autoprefixer.
- **Data Visualization:** Chart.js (v4.5.1) and vue-chartjs (v5.3.4) for dynamic business intelligence charts.
- **Frontend Utilities:** Lodash (debounce), Day.js (datetime), Axios (HTTP), Heroicons Vue 24 (Outline/Solid).
- **Build Tool:** Vite (v7.0.7) running strictly on Node.js 22 runtime engine.

1.2 Backend Framework, Database & Document Engine

- **Backend Framework:** Laravel 12 on PHP 8.2+ runtime.
- **Database Engine:** MySQL 8.x for production; SQLite support for isolated automated feature test execution.
- **Document Engine:** Barryvdh Laravel DomPDF (v3.1) for automated B2B PDF tax invoice rendering and downloads.
- **API & Security:** Laravel Sanctum (v4.0), CSRF token defense, Bcrypt password hashing, and encrypted cookies.

1.3 Authentication & Role-Based Access Control (RBAC)

Session-based authentication managed by Laravel Breeze. Authorization enforced via custom `CheckRole` middleware inspecting `users.role`:

- `super_admin` (CEO): Unconditional master access to all financial records, configuration, user deletions, and audit logs.
- `admin`: Operations administration across inventory, shipments, orders, invoices, and users.
- `sales`: Customer quote review, approval, rejection, and sales dashboard monitoring.
- `finance`: Invoices, payment processing, client credit terms, RMA returns, and financial reports.
- `purchasing`: Supplier management, Purchase Orders (PO), and goods receipt restock workflows.
- `warehouse`: Real-time inventory adjustment, shipment dispatch, and logistics courier tracking.
- `client`: Standard B2B customer; access strictly confined to public catalog, quote cart, Net-30 activations, and PDF invoices.

2. Core ERP Modules & Workflows

2.1 Core Modules

1. **Authentication & Identity:** Secure login, client registration, password resets, and user management.
2. **Public Catalog & CRM:** Paginated product exploration, multi-criteria filtering (Brand, Category, Search), and Contact Us mail routing.
3. **Sales & B2B Quotations:** Multi-item quote cart, inventory locking, quote submission, status approvals, and automated invoice conversion.
4. **Billing, Finance & Tax:** 12% VAT computation, Net-30 credit terms, PayMongo payment links, manual payments, and PDF invoicing.
5. **Inventory & Warehouse:** Live stock counts, CSV backups, automated low-stock scans, and historical stock movement audit trail.
6. **Procurement & Suppliers:** Vendor database, purchase orders, and goods receipt auto-incrementing stock.
7. **Logistics & Fulfillment:** Shipment booking, tracking generation, courier API integration, and automated email dispatches.
8. **Returns (RMA):** Itemized customer return requests on paid/credit orders for administrative review.
9. **Analytics & BI:** 6-month revenue & spend charts, top products, brand valuations, and inventory health metrics.

2.2 Business Rules & Critical Safeguards

> **Pessimistic Inventory Locking:** On quote submission, products are selected using `lockForUpdate()`. If stock is insufficient, the transaction rolls back. If valid, inventory is decremented and logged with type `sale` to prevent race conditions.

> **Automatic Reversal Handlers:** Rejecting an approved quote automatically restores deducted stock and logs an`adjustment` record. Reversing a received PO automatically deducts inventory.

> **Philippine 12% Value Added Tax:** Subtotal * 0.12 = VAT Amount. Total Amount = Subtotal + VAT Amount.

> **Net-30 B2B Credit Eligibility:** Orders qualify for Net-30 credit terms only if total aggregate item quantity? 1,000 units.

3. API Inventory & Endpoints

Method	Route URI	Allowed Roles	Status	Core Logic & Validation
GET	`/products`	Public	200 OK	Filter by category, brand, search; sort by name, brand, relevant.
POST	`/contact`	Public	302 Found	Validates required name, email, message; dispatches ContactMessageMail.
GET	`/cart`	Client (auth)	200 OK	Renders user draft quote with items and stock checks.
POST	`/cart/add`	Client (auth)	302 / 422	Enforces quantity ? 1 and quantity ? stock_quantity.
POST	`/cart/submit`	Client (auth)	302 / 404	DB transaction lock; reserves stock; status: draft ? pending.
POST	`/api/invoices/{id}/accept-terms`	Client (auth)	302 / 400 / 403	Requires total quantity ? 1000; sets status on_terms, Net-30 due date.
GET	`/invoices/{invoice}/pdf`	Client / Admin	200 OK	Generates and downloads formal B2B DomPDF tax invoice.
GET	`/admin/inventory`	Warehouse, Purchasing, Admin	200 / 403	Lists inventory, stock filter (low/out), and calculates total valuation.
GET	`/admin/inventory-export`	Warehouse, Purchasing, Admin	200 OK (CSV)	Direct CSV file export attachment of full inventory catalog.
PUT	`/admin/quotes/{quote}`	Sales, Finance, Admin	302 / 422	Status: approved/rejected. Auto-generates invoice or restores stock.
POST	`/admin/purchase-orders`	Purchasing, Admin	302 / 422	Creates draft PO with itemized quantities and unit costs.
POST	`/admin/purchase-orders/{po}/status`	Purchasing, Admin	302 / 422	Transition to received auto-increments stock and logs PO movement.
POST	`/admin/shipments`	Warehouse, Admin	302 / 422	Creates shipment; invokes LalamoveService; sends dispatch email.
POST	`/admin/invoices/{invoice}/payment`	Finance, Admin	302 / 422	Updates invoice payment state (paid/partially_paid).
POST	`/webhooks/paymongo`	PayMongo API	200 / 401	HMAC-SHA256 signature verification; marks invoice paid on event.

4. Integrations & Background Jobs

- **PayMongo Payment Gateway:** Card, GCash, Maya, and QR Ph link generation via `PayMongoService`. Built-in simulation fallback for sandbox/local runs.
- **Lalamove Logistics:** Automated courier booking simulation with dynamic tracking IDs and tracking views via `LalamoveService`.
- **Gmail SMTP Notifications:** Transactional emails powered by `ContactMessageMail`, `LowStockAlertMail`, `ShipmentDispatchedMail`, and `ShipmentDeliveredMail`.
- **Scheduled Command:** `php artisan inventory:check-stock` runs daily at 06:00 AM (Asia/Manila) to alert purchasing of items with ? 10 units.

5. Test Accounts & QA Environment Setup

Default credentials for testing in the staging/local environment (Password: `password123`):

Role	Name	Email Address	Key Testing Objectives
`super_admin`	Jubis CEO	andreapanganiban05@gmail.com	Unrestricted access, user deletions, settings cache updates.
`admin`	Jubis Admin	admin@jubismarketing.com	General administrative workflows and operational CRUD.
`sales`	Sales Department	georgeilagan62@gmail.com	Quote reviews, approvals, and stock rollback on rejections.
`finance`	Finance Department	andreabermudez0511@gmail.com	Invoice status updates, payment verifications, and reporting charts.
`purchasing`	Purchasing Department	karlammagalong6@gmail.com	Vendor directory, purchase orders, and goods receipt restocking.
`warehouse`	Warehouse & Logistics	dwyanetjhung@gmail.com	Shipment creation, courier dispatch, and stock level checks.
`client`	Demo Client	client@engineeringcorp.com	Product catalog browsing, cart submission, Net-30 credit terms.
`client`	Demo Client (Alt)	naisahaspiras24@gmail.com	Contact inquiries, client quote management, and PDF invoices.

Known QA Focus Points

- **RMA Approval Actions:** Clients can submit and Finance can view returns, but no approve/reject buttons exist in the backend.
- **Global Settings Persistence:** Settings live in Cache (`Cache::forever`); clearing cache resets to defaults.
- **Product Image Placeholders:** Over 700 Omni products without assigned images should properly show the SVG placeholder fallback.